

Infrared Frame Adventure

Make three known button presses light a tiny victory

Receive only. Unknown evidence never becomes a command.

Lesson	025 — component	Board	Arduino Mega 2560
Primary evidence	D45 lamp	Automatic ending	120 seconds
Conditional specimen	identified receiver plus owned, harmless NEC remote		
Status	host verified; physical obser- vation remains open		

The adventure

Build one receiver and one lamp. D13 announces that every pin was acquired. Then point an owned kit remote at the receiver. The first valid frame earns one long glow, the second earns two flashes, and the third unlocks three quick flashes. Unknown or damaged evidence makes a different pattern and cannot advance the count. At two minutes the sketch turns both lamps off, releases the pins, and waits for Reset.

light → pulses → frame → validity → visible reward

Numbers are evidence, not instructions. An address or command field never means “unlock”, “open”, or “launch”. This lesson does not transmit, export raw captures, replay an unknown protocol, or actuate anything beyond a resistor-limited lamp.

Specimen gate

Use the receiver only when its own marking and a primary datasheet establish its supply range, physical pin order, idle polarity, carrier band, and output level. Use only an ordinary remote you own whose harmless NEC buttons are known in advance. A similar-looking three-pin module is not an electrical identity. If either specimen is uncertain, leave the receiver unpowered and enjoy the deterministic software fixtures instead.

Part	Required identity
Mega 2560	USB-powered board
Receiver	exact active-low demodulating specimen, primary datasheet reviewed
Remote	owned ordinary remote, known harmless NEC buttons only
Visible path	ordinary LED and 1 kΩ series resistor

Wire the literal layout

Unplug USB before placing or moving anything. Match the Mega orientation, header labels, breadboard letters, and numbered rows in Plate 1.

Signal	Mega	Exact connection
TP-IR	D2	identified receiver's marked OUT function
Receiver power	5 V, GND	only when the exact datasheet permits 5 V
TP-LED	D45	row 13a; 1 k Ω from 13c to 13f; LED anode at 13h and cathode at 15h; 15j to ground rail
Ground rail	GND	rail to one Mega GND pin

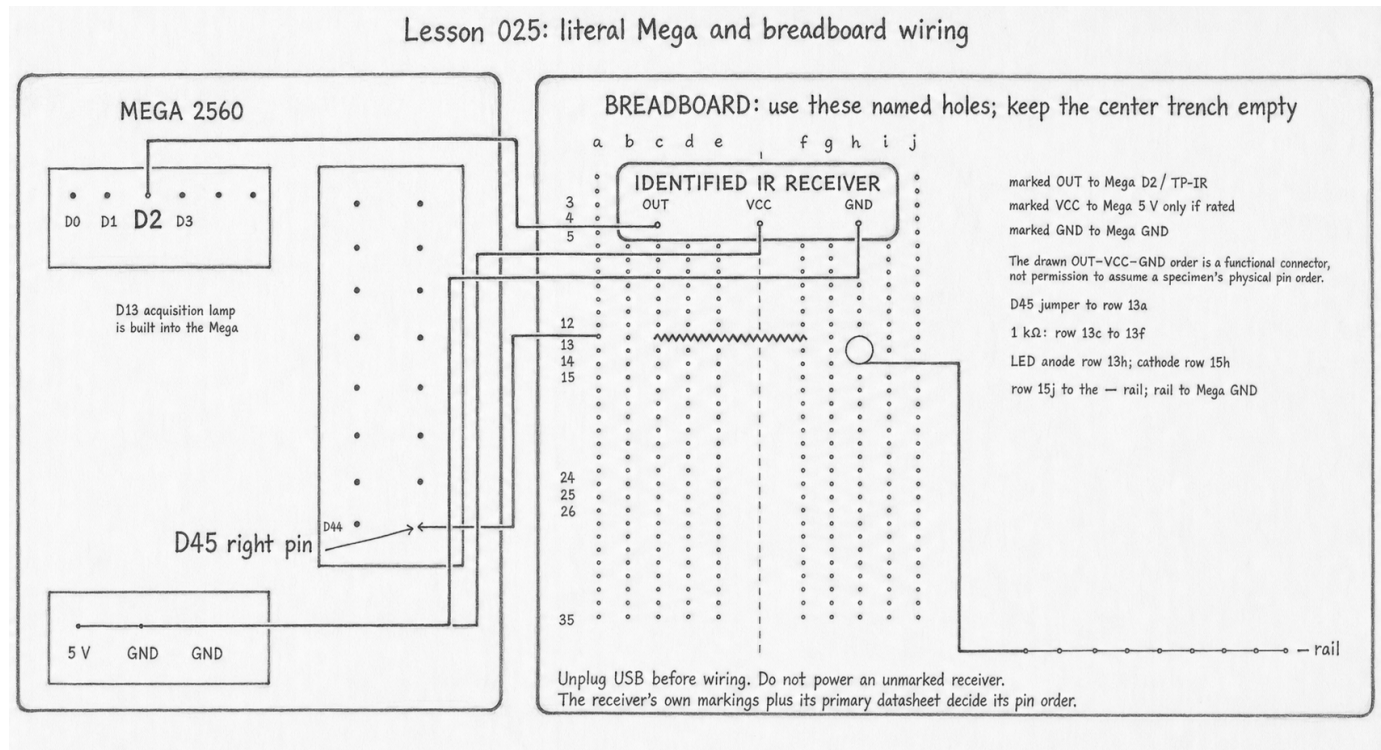


Plate 1 alternative text: literal Mega and breadboard plate. D2 connects to the identified receiver OUT function. D45 reaches breadboard row 13 through a 1 k Ω resistor and an LED spanning rows 13 and 15; the LED cathode returns through the ground rail. Receiver VCC and GND are connected only after its exact pin order is identified.

The receiver block in Plate 1 names *functions*; it deliberately does not claim that every specimen has the drawn OUT-VCC-GND physical order. Follow the exact specimen's marking and datasheet when mapping those three jumpers. The breadboard LED placement, Mega headers, and occupied holes are literal.

At an illustrative 5 V rail and 2 V LED drop, the 1 k Ω resistor limits current to about 3 mA. Unexpected heat, unstable power, a wrong resistor, or a voltage outside the rails means unplug immediately.

Watch the story unfold

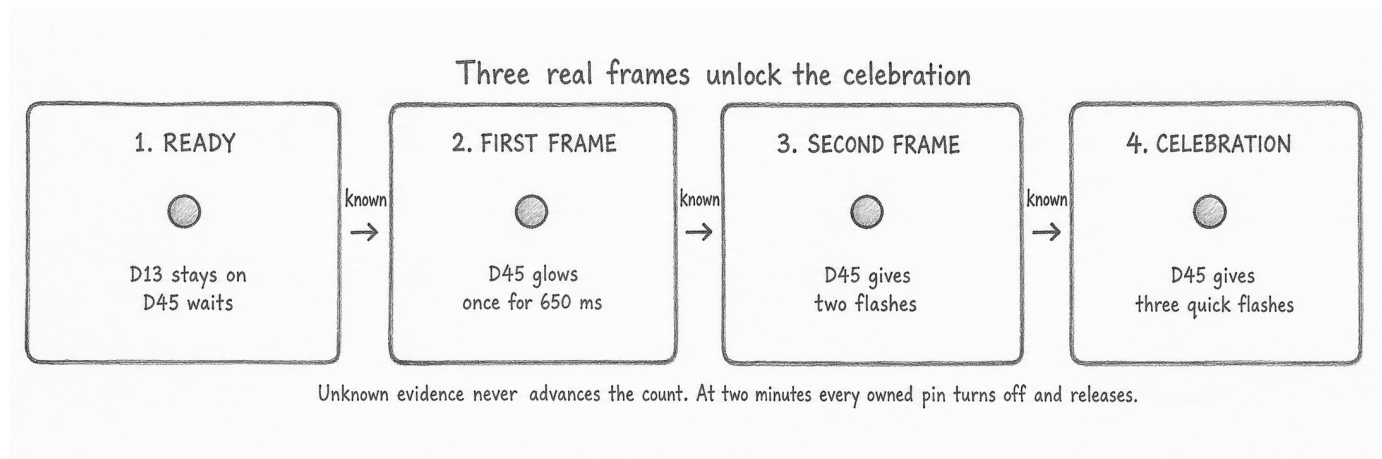


Plate 2 alternative text: four panels show D13 ready, one D45 glow after the first valid frame, two flashes after the second, and three quick flashes after the third. Unknown evidence cannot advance the story.

Stage 1: ready

After Reset, D13 stays on only if capture owns D2 and its interrupt and both lamp endpoints initialized. D45 stays dark. That visible ready state depends on successful resource acquisition; it does not claim receiver compatibility.

Stage 2: find the first frame

Press one known harmless NEC button. A valid frame makes D45 glow for 650 ms. No Serial window is needed. If D13 is on but D45 stays dark, unplug and compare D2, the receiver's exact pin order, LED polarity, resistor holes, and ground return with Plate 1.

Stage 3: unlock the progression

Press known buttons twice more, allowing each flash story to finish. The second valid frame produces two 250 ms flashes. The third and later valid frames produce three 150 ms flashes. Each later reward depends on the receiver, capture, decoder, validity decision, and earlier count all working together.

An unknown leader gives two short flashes but does not increase the valid count. A normal NEC repeat gives one short acknowledgement and also leaves the count unchanged. Timing, inverse-byte, overflow, or record trouble gives four very quick flashes. Do not seek unknown transmitters to make those patterns; automated fixtures exercise them behind the scenes.

Stage 4: discover the fixed ending

At 120 seconds, regardless of button activity, the sketch turns D45 and D13 off, shuts down both endpoints and capture, and waits. Reset starts a fresh adventure. Darkness is the circuit-native ending; it is not a claim of separately measured high impedance.

How the evidence stays honest

PulseCapture owns D2 and its interrupt. Its interrupt path stores only bounded edge timestamps. A completed **PulseFrame** remains immutable until matching acknowledgement. **InfraredDecoder** recognizes the classic 8-bit-address NEC form, validates its leader, 32 bits, final mark, and inverse bytes, and reports uncertainty without guessing.

Result	D45 story
first valid frame	one 650 ms glow
second valid frame	two 250 ms flashes
third or later valid frame	three 150 ms flashes
NEC repeat	one 150 ms acknowledgement; count unchanged
unknown protocol	two 150 ms flashes; count unchanged
timing, integrity, overflow, or record fault	four 100 ms flashes

InfraredRecordEncoder can also write one bounded canonical line:

```
IR1,<sequence>,<protocol>,<validity>,<address-hex>,<command-hex>\n
```

Serial carries that optional record, never the primary result and never raw replay material.

```
if (now.elapsedSince (adventureStarted).milliseconds () >= 120000)
{
    stopSafely ();
    return;
}

observeInfrared (adk::MicrosecondTimePoint (micros ()));
showFrameEvidence (now);
```

The automated repository suite stays behind the scenes. It covers acquisition rollback, exact timing boundaries, repeat frames, corrupt inverse bytes, overflow, stale acknowledgement, record capacity, timestamp wrap, shutdown, claim reuse, and byte-identical replay. Those checks protect the adventure without turning it into learner paperwork.

Predict, observe, interpret

Before Reset, predict that successful acquisition lights D13 while D45 remains dark. Observe D13 after setup and interpret it only as resource-acquisition evidence. Before each known press, predict the next D45 reward; observe at TP-IR and D45, then interpret agreement as evidence that the receive and decode path reached the lamp, not as general receiver compatibility. Before the two-minute deadline, predict both lamps will go dark; observe the fixed ending and interpret darkness only as the visible software ending. Pin release and high impedance remain separate, unperformed physical measurements.

If the visible story breaks

What you see	Inspect while unpowered
D13 never lights	board power, resource conflict, then initialization path
D13 on; D45 always dark	receiver identity and D2, LED orientation, rows 13/15, resistor, ground rail
every press looks unknown	exact remote protocol, receiver idle polarity, pulse timing
one short flash while holding a button	normal NEC repeat; count unchanged
four quick flashes	timing/integrity evidence or bounded-record failure
progress never reaches three	use only the known buttons; allow each frame and visible pattern to finish
lamps remain on after two minutes	unplug; fixed shutdown did not complete

Make it yours

Put paper stars numbered 1, 2, and 3 beside D45. Change only harmless presentation timing, or choose three named buttons on the owned remote and write their names on the stars. Keep all numeric fields as evidence, keep unknown evidence unable to advance, and keep the fixed two-minute ending.

Companion material. The HTML page carries searchable pin and state tables plus source links. This printable lesson carries the literal wiring plate, staged visible progression, and receive-only specimen boundary.